Name: ______________________________ Neptun code: ______________________________

(KTXFI2EBNF) Physics II. Exam 4
February 2, 2026

I. Moving Reference Frames. Special Theory of Relativity
   1. State the Galilean principle of relativity. (5 points)
   2. Write down Einstein's relation expressing mass–energy equivalence. What is its physical meaning? (5 points)

II. Limits of the Classical Conceptual Framework
   3. What is the photoelectric effect? Explain the mechanism of the photoelectric effect. (5 points)
   4. Describe the Compton experiment and provide a qualitative but precise explanation of the phenomenon of Compton scattering. (5 points)
   5. Formulate the de Broglie principle in your own words. (5 points)

III. Elements of Quantum Mechanics. Physics of Condensed Matter. Theory of Electrical Conduction
   6. What is the tunneling effect? Describe the phenomenon in your own words. (5 points)
   7. What is a forbidden band (band gap)? Define the concept. (5 points)

IV. Development of Atomic Theory
   8. Describe the Rutherford scattering experiment. (5 points)
   9. What is the Zeeman effect? Describe the phenomenon. Under what conditions does it occur and how does it manifest itself? (5 points)
   10. Given the element with atomic number 11 in the periodic table, write down its electronic configuration using the standard atomic physics notation. (5 points)

Grades: 0–25 points: Fail (1), 26–30 points: Pass (2), 31–37 points: Satisfactory (3), 38–45 points: Good (4), 46–50 points: Excellent (5).

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